

On graphs of total projective functions

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Abstract

It is well known that the graph of a total Σ_n^1 -function is Π_n^1 . We obtain the dual assertion at the third projective level: there is a model of ZFC in which the graph of every total Π_3^1 -function is Σ_3^1 . This principle is incompatible with Π_3^1 -uniformization and hence with the usual projective-determinacy picture. The construction repairs the final step of the failure-of-uniformization argument from [6]. It uses fixed binary continuations with retained origins.

1 Introduction

The projective hierarchy is proper, and this prevents any general formal duality between Σ_n^1 and Π_n^1 uniformization statements. For graphs of total functions, however, one direction behaves differently. If $f: \mathbb{R} \rightarrow \mathbb{R}$ is a total function whose graph is Σ_n^1 , then the graph of f is also Π_n^1 , since

$$(x, y) \in \text{Graph}(f) \iff \forall z((x, z) \in \text{Graph}(f) \rightarrow z = y).$$

The same argument applies in the boldface classes, with the relevant real parameters carried along. There is no analogous formal argument for total Π_n^1 -functions. The purpose of this paper is to show that at the third projective level the corresponding dual statement is nevertheless consistent.

Theorem 1.1. *Assuming $\text{Con}(\text{ZFC})$, there is a model of ZFC in which every total Π_3^1 -function has a Σ_3^1 graph.*

This is not merely a formal curiosity. As shown below, the total-graph principle obtained in Theorem 1.1 is incompatible with the Π_3^1 -uniformization property. It therefore gives another example of a projective phenomenon which cannot occur under projective determinacy.

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Uniformization, separation and reduction are among the classical regularity principles for definable sets of reals; see, for example, [15, 17, 13, 18]. Under projective determinacy, these principles are governed by the periodicity phenomena and scale machinery of the projective hierarchy; see [16, 14]. The present paper belongs to a different line of investigation: the study, by forcing, of the possible behaviour of uniformization, reduction and separation in the absence of projective determinacy. It is also connected with the older use of projective wellorderings and coding constructions to control definable sets of reals; see, for example, [1, 4, 2, 3]. In such forcing extensions the projective hierarchy can display patterns which are ruled out by the determinacy picture. Theorem 1.1 is one such pattern.

This work is part of a broader programme developed in [5, 6, 9, 8, 7, 10, 12]. The guiding theme of this programme is that forcing can produce finely calibrated projective universes in which the classical regularity principles separate from one another, combine in unexpected ways, or hold globally at levels where the determinacy intuition suggests a very different picture. The consistency of Σ_3^1 -separation was obtained in [5]; a model of Π_3^1 -reduction together with a failure of Π_3^1 -uniformization was constructed in [6]; forcing methods for Π_n^1 -uniformization were developed in [9]; and global Σ -uniformization phenomena in the presence of forcing axioms or large continuum are studied in [8, 7, 12, 11]. The present article adds another instance to this programme: it shows that total projective functions may have a definability behaviour which is not dictated by the formal duality between Σ and Π definitions and which is incompatible with the projective-determinacy pattern.

The proof is another instance of a fixed-point method for iterations of coding forcings. In these arguments one wants to iterate coding forcings long enough to meet all relevant requirements, but the correctness of a coding step depends on preservation by all future forcings which are still allowed. Thus the class of allowable continuations and the preservation requirements are defined simultaneously. This creates an apparent circularity. The way out is to organize the construction as a transfinite derivative on classes of allowable iterations and to work at the eventual fixed point. This method is central in the Π_3^1 -reduction construction of [6] and in the forcing construction for Π_n^1 -uniformization in [9]; a simpler version also appears in the forcing construction for Σ_3^1 -separation in [5]. The technique has recently been extended and refined in [11].

Here the fixed-point method is adapted to the total-graph problem. One hierarchy is computed in W , using W -names and retained origins. A later continuation is an inherited quotient. An old value cannot become protected merely because the base of a derivative has changed. If a preservation witness appears after an empty-origin action, common-prefix squaring gives two values and hence non-functionality. The extension argument selects a witness at its first nonempty preservation rank.

The coding machinery is from [6]. Its final extraction of a non-uniformizable relation used an assertion about partial graphs which cannot hold. Section 4 proves cone transport and closure under fixed binary continuations. Dense destruction tasks are scheduled separately, so their lengths need no common count-

able bound. Section 5 then proves the total-graph implication.

2 Preliminary considerations

We first record why Theorem 1.1 implies a failure of Π_3^1 -uniformization.

Lemma 2.1. *In ZFC there is a total Π_3^1 relation $R \subseteq \mathbb{R}^2$ with $\text{dom}(R) = \mathbb{R}$ which cannot be uniformized by any function whose graph is Σ_3^1 .*

Proof. Let $B \subseteq \mathbb{R}$ be a Σ_3^1 set which is not Δ_3^1 in the boldface sense. Fix a Π_2^1 relation P such that

$$x \in B \iff \exists y P(x, y).$$

By applying a recursive bijection if necessary, we may assume without loss of generality that

$$\forall x \forall y (P(x, y) \rightarrow y \neq 0).$$

Define $R \subseteq \mathbb{R}^2$ by

$$R(x, y) \iff P(x, y) \vee (x \notin B \wedge y = 0).$$

Then R has full first projection. Moreover, R is Π_3^1 , since projective pointclasses are closed under finite unions and P is $\Pi_2^1 \subseteq \Pi_3^1$, while the second disjunct is Π_3^1 .

Suppose for contradiction that $g: \mathbb{R} \rightarrow \mathbb{R}$ uniformizes R and that $\text{Graph}(g) \in \Sigma_3^1$. Since g is total, its graph is also Π_3^1 , hence $\text{Graph}(g) \in \Delta_3^1$. For every $x \in \mathbb{R}$, since g uniformizes R , $R(x, g(x))$ holds. If $x \notin B$, then $\neg \exists y P(x, y)$, so the second disjunct of R must hold and $g(x) = 0$. Conversely, if $x \in B$, then $P(x, g(x))$ holds; by the choice of P , this implies $g(x) \neq 0$. Thus

$$x \notin B \iff g(x) = 0.$$

Because $\text{Graph}(g) \in \Delta_3^1$, the preimage $g^{-1}[\{0\}]$ is Δ_3^1 . This implies $\mathbb{R} \setminus B \in \Delta_3^1$, contradicting the choice of B . $\square$

Corollary 2.2. *In every model in which all total Π_3^1 -functions have Σ_3^1 graphs, the Π_3^1 -uniformization property fails.*

Proof. If Π_3^1 -uniformization held, the relation R from Lemma 2.1 would be uniformized by some function f_R such that $\text{Graph}(f_R) \in \Pi_3^1$. By the assumed property, $\text{Graph}(f_R)$ would also be Σ_3^1 , contradicting Lemma 2.1. $\square$

Remark 2.3. The elementary observation used in the introduction has a corresponding dual use: whenever a universe satisfies the appropriate projective uniformization principle, a total Σ_n^1 relation can be uniformized by a function whose graph is both Σ_n^1 and Π_n^1 . We shall not use this observation below.

3 The base model and coding machinery

The construction uses the coding machinery developed in [6]. We recall its preparation, coding forcings and no-accidental-codes lemma. The derivative and continuation arguments needed here are stated separately in Section 4.

We work over a ground model W obtained by an ω -distributive forcing over L . Starting in L , let

$$\mathbb{Q} = \mathbb{Q}^0 * \mathbb{Q}^1 * \mathbb{Q}^2$$

be the standard three-step forcing from [6]. The first iterand adds, by countable approximations, a sequence

$$\vec{S} = \langle S_\alpha \mid \alpha < \omega_1 \rangle$$

of pairwise independent Suslin trees over L . The second iterand adds a branch b_α through every S_α ; the third codes the trees into the stationary-set pattern. The sequence is independent in $L[G^0 * G^2]$, not in W . The forcing $\mathbb{Q}$ is ω -distributive, hence adds no reals. If $G \subseteq \mathbb{Q}$ is L -generic and $W = L[G]$, then in W the sequence $\vec{S}$ is uniquely characterized by a Σ_1 formula over $H(\omega_2)$ with parameter ω_1 . We use this definition in the allowable extensions considered below, where the defining pattern is preserved. Thus, throughout the later forcing construction, the phrase “the α -th coding tree” continues to have the same meaning in every relevant intermediate extension.

Because the final theorem is a boldface statement, real parameters must be included in the coding. We fix a recursive enumeration

$$\langle \varphi_m(v_0, v_1, v_2) \mid m < \omega \rangle$$

of all lightface Π_3^1 formulas in three displayed real variables. For a real parameter p , set

$$A_{m,p} = \{(x, y) \in \mathbb{R}^2 \mid \varphi_m(x, y, p)\}.$$

Thus every boldface Π_3^1 subset of the plane is some $A_{m,p}$. Given reals p, x, y and $m < \omega$, let (p, x, y, m) denote a fixed recursive real code for the quadruple.

We use the same local coding device as in [6]. The forcing which writes the real (p, x, y, m) into the Suslin-tree pattern is denoted by

$$\text{Code}(p, x, y, m) := \mathbb{P}_{(p,x,y,m)} = \mathbb{C}(\omega_1)^L * \dot{\mathbb{A}}(\dot{Y}).$$

Here $\mathbb{C}(\omega_1)^L$ is the fixed constructible forcing with countable conditions. Fix the constructible coding map ρ from [6, Section 3.2.1]. A reservoir generic g determines

$$h_g = \{\rho(g \cap \alpha) : \alpha < \omega_1\}.$$

For the real t coding (p, x, y, m) , the coding set selects the branches

$$b_{\omega\gamma+2n+t(n)} \quad (\gamma \in h_g, n < \omega)$$

already present in W , together with the club witnesses needed to reconstruct the trees. The reshaped set Y and the Jensen–Solovay forcing $\dot{\mathbb{A}}(Y)$ make this

information available in $L[r]$, where r is the coding real. The coding stage does not add branches to the trees.

Reservoir and almost-disjoint forcing coordinates have disjoint labels. The sets h_g of tree indices are almost disjoint, rather than disjoint. For a countable family of independent reservoirs, their pairwise intersections are bounded below ω_1 . Freshness means a new independent reservoir coordinate. We retain the original names for Y under restrictions and amalgamation; we do not recompute the coding recipe in a larger universe. The local models used by the projective decoder remain part of that recipe.

Once Y is present, $\mathbb{A}(Y)$ has the same finite conditions and order in every further extension with the same ω_1 . The predicate decoding an introduced tuple is upward persistent. Exact decoding for the whole construction is the no-accidental-codes assertion of [6, Lemma 3.4], not a consequence of disjoint tree blocks.

We first record the one-step coding consequence. The corresponding iteration facts are collected below, after the definition of allowable forcings.

Lemma 3.1. *There is a recursive coding of quadruples $t = (p, x, y, m)$ by reals and a uniformly Σ_3^1 predicate*

$$\Phi(p, x, y, m)$$

with the following one-step property. Let M be an intermediate extension of an allowable W -presentation. If $t = (p, x, y, m) \in M$ and one forces over M with a single coding forcing $\text{Code}(p, x, y, m)$, using a fresh independent reservoir, then the resulting extension satisfies

$$\Phi(p, x, y, m).$$

The resulting coding real decodes t . In an allowable construction no different tuple is decoded unless it is explicitly coded. Once $\Phi(t)$ holds, it remains true in further allowable extensions.

Proof. Apply the coding construction and the no-accidental-codes lemma of [6, Section 3.2.1 and Lemma 3.4] to the recursive code of t . The latter lemma uses the branch-factor argument in the associated inner models. For persistence, write $\Phi(t)$ as $\exists r \psi(r, t)$, where ψ is Π_2^1 , and apply Shoenfield absoluteness to the witness r . $\square$

We next recall the class of allowable forcings from [6]. In the present paper it is useful to formulate the definition so that the product/iteration hybrid is visible. At each stage the construction first adds a fresh copy of $\mathbb{C}(\omega_1)^L$ on a new reservoir coordinate, independently of the generic object already constructed. Only after this independent reservoir coordinate has been added does the bookkeeping split into two cases: either the stage is padding and the second component is trivial, or the stage codes a real named in the current intermediate extension by Jensen–Solovay almost-disjoint coding. Thus the object handled by the bookkeeping is a real, not an arbitrary set $Y \subseteq \omega_1$; the set Y used by

the almost-disjoint forcing is produced uniformly from that real and from the fresh reservoir generic.

Definition 3.2. *Let $\alpha < \omega_1$. A hybrid allowable presentation of length α over W consists of a sequence*

$$\langle \mathbb{P}_\beta \mid \beta \leq \alpha \rangle,$$

a bookkeeping function $F \in W$ with domain α , pairwise disjoint reservoir coordinates, and the associated almost disjoint coding areas, satisfying the following clauses.

First, $\mathbb{P}_0$ is trivial. At a successor stage $\beta + 1$, let

$$\mathbb{C}_\beta = \mathbb{C}(\omega_1)^L$$

be the fresh reservoir forcing assigned to β . This is not a $\mathbb{P}_\beta$ -name and is not interpreted inside the current $\mathbb{P}_\beta$ -generic extension. The next stage is obtained by first taking the independent product with this fresh reservoir coordinate and then applying a second, iteration-like component:

$$\mathbb{P}_{\beta+1} = (\mathbb{P}_\beta \times \mathbb{C}_\beta) * \dot{\mathbb{B}}_\beta.$$

Here $\dot{\mathbb{B}}_\beta$ is a $(\mathbb{P}_\beta \times \mathbb{C}_\beta)$ -name determined by $F(\beta)$ as follows.

(1) Padding stage. *If $F(\beta) = \star$, then*

$$\dot{\mathbb{B}}_\beta = \dot{\mathbf{1}}.$$

Thus the stage still adds the fresh reservoir generic for $\mathbb{C}_\beta$, but it performs no almost-disjoint coding.

(2) Coding stage. *If $F(\beta)$ is a code for a $\mathbb{P}_\beta$ -name $\dot{r}_\beta$ such that*

$$\mathbb{P}_\beta \Vdash \dot{r}_\beta \in 2^\omega,$$

then, after forcing with $\mathbb{P}_\beta \times \mathbb{C}_\beta$, the real

$$r_\beta = (\dot{r}_\beta)^{G_\beta}$$

is read from the current $\mathbb{P}_\beta$ -generic object G_β , while the reservoir generic $c_\beta \subseteq \mathbb{C}_\beta$ is independent of G_β . In the extension by $G_\beta \times c_\beta$, let $Y_\beta \subseteq \omega_1$ be the standard reshaped set computed uniformly from r_β , from c_β , and from the coding area determined by that reservoir. Then

$$\dot{\mathbb{B}}_\beta = \dot{\mathbb{A}}(\dot{Y}_\beta),$$

where $\mathbb{A}(Y_\beta)$ is the Jensen–Solovay almost-disjoint coding forcing, using the fixed constructible almost-disjoint family, which codes Y_β into a real.

At limit stages the support convention is mixed: the reservoir coordinates $\mathbb{C}_\beta$ are taken with countable support, while the almost-disjoint coding components $\dot{\mathbb{B}}_\beta$ are taken with finite support. Reservoir products and their countable supports are

formed in W and retained in every quotient. Freshness of reservoir coordinates is part of the definition.

A forcing admitting such a presentation is called allowable, or 0-allowable. If $\dot{r}_\beta$ is the canonical real code for a quadruple (p, x, y, m) , we write the corresponding coding stage as

$$\text{Code}(p, x, y, m).$$

This notation abbreviates the whole stage operation over $\mathbb{P}_\beta$: first adjoining the independent reservoir coordinate $\mathbb{C}_\beta$, and then performing the almost-disjoint coding determined by the reshaped set associated with the real coding (p, x, y, m) .

The definition should be read as saying that the construction iteratively codes reals

$$r_\beta = (\dot{r}_\beta)^{G_\beta}$$

whenever the bookkeeping presents such a name. The set $Y_\beta \subseteq \omega_1$ is merely the internal reshaped object which makes the code projectively recognizable. In particular, the bookkeeping does not choose an arbitrary $Y \subseteq \omega_1$; it chooses, or skips, a real to be coded, and Y_β is then produced by the fixed coding recipe.

The important point is that the fresh reservoir coordinate is present in both cases. A padding stage is

$$\mathbb{P}_\beta \mapsto (\mathbb{P}_\beta \times \mathbb{C}_\beta) * \mathbf{1},$$

whereas a coding stage is

$$\mathbb{P}_\beta \mapsto (\mathbb{P}_\beta \times \mathbb{C}_\beta) * \mathbb{A}(Y_\beta).$$

Thus the symbol $\dot{\mathbb{B}}_\beta$ denotes only the second, iteration-like component of the stage. The Cohen reservoir forcing $\mathbb{C}_\beta$ is never merely a $\mathbb{P}_\beta$ -name for the next iterand; it is an independent product coordinate added before the stage decision is implemented. This is the product characteristic of the construction. The dependence of Y_β on the current real r_β , and the fact that later bookkeeping may depend on earlier coding reals, is the iteration characteristic.

Equivalently, for any fixed length $\delta < \omega_1$, one may first add by a countable-support product all reservoir generics

$$\left(\prod_{\beta < \delta} \mathbb{C}_\beta \right)_{\text{c.s.}}$$

and then perform the finite-support iteration of the almost-disjoint coding components, using the reservoir generic assigned to the relevant stage. This front-loaded presentation is canonically isomorphic to the hybrid presentation, provided the bookkeeping names are interpreted with the same regular supports. It is often useful conceptually, because it shows that the Cohen part is a product of independent reservoirs, while all genuine coding decisions occur in the almost-disjoint coding part.

Remark 3.3. The mixed presentation causes no additional preservation or definability difficulty. Indeed, any allowable forcing of length $\delta < \omega_1$ can be rearranged, up to the canonical isomorphism supplied by the chosen disjoint coordinates, as

$$\left(\prod_{\beta < \delta} \mathbb{C}_\beta \right)_{\text{c.s.}} * \left(\prod_{\beta < \delta}^* \dot{\mathbb{B}}_\beta \right)_{\text{f.s.}},$$

where each $\mathbb{C}_\beta$ is a fresh copy of $\mathbb{C}(\omega_1)^L$, and where $\dot{\mathbb{B}}_\beta$ is either the trivial forcing or the almost-disjoint coding forcing $\mathbb{A}(Y_\beta)$ computed from the real named at stage β and from the reservoir generic c_β . The first factor is σ -closed, since it is a countable-support product of $< \omega_1$ many copies of $\mathbb{C}(\omega_1)^L$. The second factor is a finite-support iteration of ccc almost-disjoint coding forcings. Hence ω_1 is preserved.

The rearrangement does not change the intended codes. Moving all reservoir coordinates to the front merely supplies the independent generics c_β in advance. The later finite-support iteration still computes the names $\dot{r}_\beta$ in the same intermediate extensions and then applies the same uniform reshaping and almost-disjoint coding recipe. Thus the hybrid notation is only a bookkeeping convenience, not an additional forcing principle.

If two allowable forcings use disjoint reservoir and almost-disjoint forcing coordinates, their product interleaves the two lists of stages while retaining their names. The reservoir product is formed in W and is σ -closed there; the subsequent finite-support coding iteration is ccc. We do not assert that $\mathbb{C}(\omega_1)^L$ is σ -closed after reals have been added, or replace a quotient by a newly computed countable-support product. Exact decoding uses [6, Lemma 3.4].

Lemma 3.4. *The following hold for allowable forcings over W :*

- (i) *If $\mathbb{P}$ is an allowable hybrid presentation of length δ , then at every stage $\beta < \delta$ the fresh reservoir coordinate $\mathbb{C}_\beta$ has size $\aleph_1$, and*

$$\mathbb{P}_\beta \times \mathbb{C}_\beta \Vdash |\dot{\mathbb{B}}_\beta| \leq \aleph_1,$$

where $\dot{\mathbb{B}}_\beta$ is the second, iteration-like component of the stage.

- (ii) *Every allowable forcing over W preserves ω_1 .*
- (iii) *The product of two allowable forcings with disjoint reservoir coordinates and disjoint coding coordinates is allowable.*
- (iv) *If $\langle \mathbb{P}_\alpha \mid \alpha \leq \omega_1 \rangle \in W$ is an ω_1 -length hybrid construction such that each countable initial segment is allowable over W , then $\mathbb{P}_{\omega_1} \Vdash \text{CH}$.*
- (v) *If a stage of an allowable construction explicitly performs the coding operation $\text{Code}(p, x, y, m)$, using a fresh reservoir coordinate, then the next intermediate extension satisfies $\Phi(p, x, y, m)$.*

- (vi) *In any allowable construction satisfying the coordinate convention, apart from codes already present in the initial model, the tuples decoded in the final extension are exactly the tuples explicitly written by stages of the construction.*
- (vii) *Once $\Phi(t)$ holds at some stage, it remains true in all later allowable extensions which use fresh reservoir coordinates. We call this persistence in what follows.*
- (viii) *In any countable or ω_1 -length construction satisfying the coordinate convention and having allowable countable initial segments, if $\Phi(t)$ holds in the final model and was not already true in the initial model, then there is a least stage at which $\text{Code}(t)$ was explicitly introduced. In W no tuple satisfies the decoding predicate, by the same no-accidental-codes argument. Thus every final instance of $\Phi(t)$ has such a first stage.*

Proof. For (i), the independent reservoir coordinate is a fresh copy of $\mathbb{C}(\omega_1)^L$, hence has size $\aleph_1$. After the product with this coordinate has been taken, the second component is either trivial or is Jensen–Solovay almost-disjoint coding with respect to the fixed almost-disjoint family of size $\aleph_1$. Hence the displayed size bound holds in the relevant intermediate model.

For (ii), use the rearrangement from Remark 3.3. The reservoir part is a countable-support product of $< \omega_1$ many copies of $\mathbb{C}(\omega_1)^L$, hence is σ -closed. After forcing with it, the almost-disjoint coding part is a finite-support iteration of ccc forcings of length $< \omega_1$, hence is ccc. Therefore the whole allowable forcing preserves ω_1 .

For (iii), choose disjoint reservoir and coding coordinates for the two forcings. Interleave their bookkeeping functions. The rearranged form of the product is again a countable-support product of independent $\mathbb{C}(\omega_1)^L$ reservoirs followed by a finite-support iteration of almost-disjoint coding forcings, with trivial second components at padding stages. Since the coordinates are disjoint and the definition of $\mathbb{A}(Y)$ is absolute once Y is present, this interleaving is an allowable forcing and is canonically isomorphic to the product.

For (iv), frontload the inherited reservoir product in W . It is σ -closed and preserves CH. The subsequent finite-support coding iteration is ccc and has a dense set of size at most ω_1 . It has at most $(\omega_1)^{\aleph_0} = \omega_1$ nice real names over the reservoir extension. Thus the final model satisfies CH. The real-localization argument of Lemma 4.16 justifies the use of countable prefixes below.

Clause (v) is the one-step coding lemma. Clauses (vi) and (viii) use [6, Lemma 3.4], applied to the retained allowable presentation. At the ω_1 limit, a real witness to $\Phi(t)$ appears in a countable initial segment; Shoenfield absoluteness gives the same decoding there. Clause (vii) follows from the Σ_3^1 form of Φ as above. The absence of codes in W concerns the decoder, not the absence of branches: W already contains all the branches b_α . $\square$

4 The derivative operator and ∞ -allowable forcings

We compute one hierarchy over W . Regular supports are used to evaluate W -names; they are not bases for new derivative computations.

Definition 4.1. *Let $\mathbb{P}$ be an allowable W -presentation. A set A of its coordinates is a regular support if it contains the dependencies of all retained names and action records, and its inherited forcing satisfies*

$$\mathbb{P}_A < \mathbb{P}.$$

If G is $\mathbb{P}$ -generic, put $G_A = G \cap \mathbb{P}_A$. Initial segments are regular supports. Every retained coding action keeps its origin presentation, origin conditions, parameter names, and coded real name. These records are transported under embeddings, rather than replaced by the ambient preceding forcing.

Bookkeeping is exhaustive if every required tuple of reals appearing in an initial segment is considered cofinally often. We choose names for ω_1 -enumerations of those reals and list tuples of indices; we do not list every syntactically different name.

We use the following presentations. A template has a fixed countable skeleton. Its reservoir part is the inherited W -product

$$C_\delta = \left(\prod_{\nu < \delta} \mathbb{C}(\omega_1)^L \right)_{\text{c.s.}} \upharpoonright \vec{s}, \quad \vec{s} \in C_\delta \cap W.$$

Its coding part has finite support. On a prefix Boolean cone a factor is $\mathbb{A}(\dot{Y})$, possibly below a condition name; outside its active cone it is 1. The original coding recipe supplies $\dot{Y}$. Reservoir roots are fixed in W , including at inactive positions.

On an active prefix cone q , an action record consists of

$$(A, a, \dot{p}, \dot{x}, \dot{w}, m; c, e), \quad 0 < c \leq a, \quad e : \text{RO}(\mathbb{P}_A) \upharpoonright c \longrightarrow \text{RO}(\mathbb{P}_{\text{prefix}}) \upharpoonright q.$$

Here e is unital and regular. The first six entries are immutable. The current domain c , map e and active cone may change. In particular, a restriction does not replace A by $A \upharpoonright c$. Origins and name dependencies have well-founded reference lists.

Permitted maps identify dependency coordinates and the original names on their active cones. They are generated by origin-closed coordinate inclusions, relabelings, common-prefix branch inclusions, restriction to Boolean cones, the binary construction below, and composition. We do not quantify over arbitrary complete subalgebras. A cone is a marked restriction of its template. It retains the template's record list, including dormant records; the derivative still tests their original (A, a) .

Normalize dense condition labels into ω_1 . The resulting codes form a set $\mathcal{E} \subseteq H_{\omega_2}^W$, with the coding apparatus fixed. Conditions, names, Boolean values, antichain records and maps on dense posets have codes of hereditary size at most ω_1 . Ranks are auxiliary data. Normalization fixes the physical trees and the map ρ . Set $\Gamma_0 = \mathcal{E}$. All recursions and forcing relations below are in W . We identify generic filters with their extensions to Boolean completions.

Definition 4.2. Let $A \in \Gamma_\alpha$, let $a \in \text{RO}(\mathbb{P}_A)^+$, and let $\dot{p}, \dot{x}, \dot{y}$ be $\mathbb{P}_A$ -names. Write

$$\text{Pres}_\alpha(A, a; \dot{p}, \dot{x}, \dot{y}, m)$$

if, for every $0 < c \leq a$, every $R \in \Gamma_\alpha$, and every permitted presentation-respecting regular embedding of origin cones

$$j : \text{RO}(\mathbb{P}_A) \restriction c \longrightarrow \text{RO}(\mathbb{P}_R) \restriction r,$$

one has

$$r \Vdash_{\mathbb{P}_R} \varphi_m(j\dot{x}, j\dot{y}, j\dot{p}).$$

The names are restricted to the domain cone before transport. Identity cones are included. Embeddings fix the origin's forcing data and dependency order. In a $\mathbb{P}_A$ -extension define

$$Y_{A, G_A, \dot{x}, \dot{p}, m}^\alpha = \{\dot{y}^{G_A} : \exists a \in G_A \text{ Pres}_\alpha(A, a; \dot{p}, \dot{x}, \dot{y}, m)\}.$$

We abbreviate this set by Y_A^α when the names are fixed. It is the evaluation of a W -name, not a derivative computed in $W[G_A]$. A fixed name for an ω_1 -enumeration of the origin reals gives a witness-set name of size at most ω_1 .

Definition 4.3. A decorated presentation P belongs to $\Gamma_{\alpha+1}$ if $P \in \Gamma_\alpha$, its retained origins belong to Γ_α , and each coding action, with recorded origin (A, a) and coded name $\dot{w}$, satisfies

$$a \Vdash_{\mathbb{P}_A} (Y_A^\alpha = \emptyset) \vee (\dot{w} \in Y_A^\alpha).$$

Thus an empty preservation set permits any guess; a nonempty set requires a preservation witness. A selected witness is recorded, but need not be the least witness in a wellorder of coordinate labels. Padding has no coding test. At a limit ordinal λ put

$$\Gamma_\lambda = \bigcap_{\alpha < \lambda} \Gamma_\alpha.$$

The intersection uses the same decorated presentations.

Lemma 4.4. For every α , membership in Γ_α is definable in W , uniformly in the presentation and α .

Proof. Use set recursion on $\mathcal{E}$, quantifying over the specified embeddings and names. At limit stages take intersections. $\square$

Lemma 4.5. *If $\alpha < \beta$, then $\Gamma_\beta \subseteq \Gamma_\alpha$. For an origin admitted at both levels, $\text{Pres}_\alpha \Rightarrow \text{Pres}_\beta$ and $Y_A^\alpha \subseteq Y_A^\beta$.*

Proof. The first assertion is part of the recursion. The second follows by restricting the set of extensions quantified over. $\square$

Lemma 4.6. *Let P be a designated regular dependency support of allowable presentations Q_0, Q_1 , with independent coordinates outside P . Then*

$$S = Q_0 \otimes_P Q_1 = \mathbb{P}_P * ((\mathbb{P}_{Q_0}/\dot{G}_P) \times (\mathbb{P}_{Q_1}/\dot{G}_P))$$

has an allowable W -presentation. Both branch embeddings are regular and agree on P . All names and origin records are retained.

Proof. Frontload the reservoir product in W . Interleave the two coding orders, retaining each branch's dependencies. For the fixed constructible family $\{d_\xi : \xi < \omega_1\}$, the conditions of $\mathbb{A}(Y)$ are

$$(s, F) \in [\omega]^{<\omega} \times [\{d_\xi : \xi < \omega_1\}]^{<\omega},$$

with

$$(t, H) \leq (s, F) \iff s \subseteq t \wedge F \subseteq H \wedge \forall d_\xi \in F (\xi \in Y \Rightarrow t \cap d_\xi = s \cap d_\xi).$$

These conditions and this order are unchanged after adjoining the other branch, since the original Y -name is retained. Hence a branch coding factor commutes with an unrelated factor. Restriction of reservoir conditions and finite coding supports gives the branch projections. Conversely, union of compatible reservoir conditions and finite coding supports gives the dense map from the interleaved presentation to the displayed amalgam, on the dense presentation where each finite almost-disjoint condition name is decided. Countable reservoir supports and finite coding supports are preserved. This also proves regularity of the branch embeddings. No tree-index permutation or new branch forcing is used. $\square$

Lemma 4.7. *The presentations above admit condition restrictions with unchanged origin tests. They preserve ω_1 and CH, and have exact decoding.*

Proof. Let $e : B \restriction c_0 \rightarrow C \restriction q_0$ be a unital regular embedding and $0 < d \leq q_0$. Put

$$c = \pi_e(d) = \bigwedge \{b \leq c_0 : d \leq e(b)\}, \quad e_d(b) = e(b) \wedge d \quad (b \leq c).$$

Then $c > 0$ and $e_d : B \restriction c \rightarrow C \restriction d$ is unital, complete and injective. Indeed, if $0 < b \leq c$ and $e(b) \wedge d = 0$, then $d \leq e(c_0 \setminus b)$ and $c \leq c_0 \setminus b$, a contradiction. Restricted names have the same interpretations. Applying this formula to each active record changes only its map and domain. Its original origin, condition and names remain fixed. Dormant records are retained.

For a reservoir condition s , a constructible bijection $f_s : \omega_1 \rightarrow \omega_1 \setminus \text{dom}(s)$ gives

$$u \mapsto s \cup \{(f_s(\xi), u(\xi)) : \xi \in \text{dom}(u)\}.$$

This is an isomorphism onto $\mathbb{C}(\omega_1)^L \upharpoonright s$. For a fixed W -array of roots use these maps coordinatewise in the inherited W -product.

For $q = (s, F) \in \mathbb{A}(Y)$ put

$$D_q = \bigcup \{d_\xi \in F : \xi \in Y\}, \quad f_q(u, H) = (s \cup (u \setminus D_q), H \cup F).$$

This is a complete projection onto $\mathbb{A}(Y) \upharpoonright q$. If $(t, K) \leq f_q(u, H)$, then

$$(u \cup (t \setminus s), K) \leq (u, H), \quad f_q(u \cup (t \setminus s), K) = (t, K).$$

The first inequality uses the restraints in H ; the second uses $(t \setminus s) \cap D_q = \emptyset$. On the cone below q , f_q is the identity. The projected generic real is

$$r_q = s \cup (r \setminus D_q), \quad |r_q \triangle r| < \omega.$$

It codes the same Y . All formulas use the original Y -name.

These projections iterate with finite support: lift the finitely many coordinates of a condition in dependency order. At a limit take their union. The master reservoir product is unchanged. Thus a coned presentation has a complete projection from a presentation with full almost-disjoint factors. Its active tuple names are pulled back through the projection.

To prove exact decoding, fix a putative decoded tuple t with formula index m . In this full presentation, code the same tuple at each active stage. At every inactive stage code a fixed constructible dummy tuple with index $m' \neq m$, using that stage's reservoir. Project its almost-disjoint factor to 1. The full presentation is a restriction of an ordinary allowable presentation, in the sense of [6, Definition 3.1], below its one fixed countable reservoir root array. A generic extension of the coned model by the projection quotient is therefore an ordinary allowable model whose codes are the original active codes and these dummy codes. If $\Phi(t)$ held in the coned model, it persists to that extension. By [6, Lemma 3.4], t must have been explicitly coded there. Its index excludes every dummy code, so it was an active original code. Conversely, the one-step lemma and persistence decode each active code. The same argument applies to each reduction decoder.

The reservoir product is σ -closed in W ; the coding iteration is ccc. Hence ω_1 and CH are preserved, as in Lemma 3.4. At an ω_1 limit, Lemma 4.16 puts a real decoding witness in a countable prefix, and Shoenfield absoluteness reduces the assertion to that prefix. The projection f_q is used only for this forcing envelope, not as a permitted origin map. $\square$

Lemma 4.8. *Let A be a common dependency origin of U, R , on matching source cones. Let $r \in R$ have nonzero origin projection b . Fix a U -cone $d \leq j_U(b)$ and put*

$$b_d = \pi_{j_U}(d), \quad r_d = r \wedge j_R(b_d).$$

There is a regular extension of U which, on d , adjoins $R \upharpoonright r_d$ over $A \upharpoonright b_d$ and, on $\neg d$, pads the same countable skeleton. All origin records are retained.

Proof. Move the dependency support to the beginning. Apply Lemma 4.6. Both d and r_d have origin projection b_d , so their conjunction in this amalgam is nonzero. The branch maps restricted by Lemma 4.7 are regular.

Use one fixed copy of the reservoir array outside A . Impose the same fixed W -roots from r on both d and $\neg d$. On d use the original coding factors with the inherited restriction r_d ; on $\neg d$ use 1. Mix their factor names over this binary partition. Lemma 4.7 supplies the full coding envelope. The two U -branch maps agree on their partition and join to a unital regular embedding of U . An R -origin map is required only on its active cone and is composed with the restricted branch map. Its original (A', a') is unchanged.

Use conditions with decided finite almost-disjoint entries. They are dense: deciding an entry introduces finitely many earlier entries. Their indices decrease along each recursion branch. The finitely branching well-founded recursion tree is finite. Restriction of such a condition to an origin-closed support is an actual origin condition and is a strong reduction. Thus the fixed roots outside the origin form one W -array. No array of roots is chosen from an earlier generic. $\square$

Lemma 4.9. *For every α , Γ_α is closed under marked cones, origin-closed regular supports, relabelings, common-prefix amalgams, fixed binary continuations and W -specified countable concatenations of these operations. Copies retain their original tests. If P is admitted and G is generic, its inherited continuations are*

$$\Gamma_\alpha[P, G] = \{\mathbb{P}_R/j[G] : R \in \Gamma_\alpha, j : \mathbb{P}_P \rightarrow \mathbb{P}_R \text{ permitted}\},$$

with their quotient cones. This is not allowability recomputed over $W[G]$.

Proof. Induct on α . At level zero use Lemmas 4.6, 4.7 and 4.8. Retained coordinate supports are regular by restriction and union of finite coding conditions and reservoir conditions. A countable concatenation of fixed countable skeletons has countable ordinal length and a fixed countable reservoir array in W .

At a successor level the output belongs to Γ_α by induction. Each output record comes from an input record and retains exactly

$$a \Vdash_A (Y_A^\alpha = \emptyset) \vee (\dot{w} \in Y_A^\alpha).$$

Only its current map or activity changed. Its origin A was already in Γ_α . Padding introduces no test. This includes dormant records of marked cones, so restriction cannot remove an origin test. Hence the output belongs to $\Gamma_{\alpha+1}$.

At a limit use the same output code at every lower rank and take the intersection. Isomorphisms and their inverses transport permitted maps, so they transport the preservation predicates as well. Quotients retain the code and records of their original W -presentations. $\square$

Lemma 4.10. *The sequence $\langle \Gamma_\alpha : \alpha \in \text{Ord} \rangle$ stabilizes in W . Its stable value is nonempty.*

Proof. It is a decreasing sequence of subsets of the fixed set $\mathcal{E}$. If every successor inclusion were strict, distinct presentations would be removed at more than $|\mathcal{E}|$ stages. Once two consecutive classes coincide the successor operator, which has no additional stage parameter, keeps them equal. The trivial and padding presentations survive all levels. $\square$

Definition 4.11. Let θ be the first stabilization stage and put $\Gamma_\infty = \Gamma_\theta$. An ∞ -allowable continuation means an inherited member of $\Gamma_\infty[P, G]$, with its origin records. At the stable level, I records actions with $Y_A^\infty \neq \emptyset$, J records actions with $Y_A^\infty = \emptyset$, and D records pairs of distinct preservation witnesses for the same (m, p, x) . These classifications are derived from the retained action record; an action is not permanently labelled a guess at every rank where the preservation set was empty.

Lemma 4.12. Let $P \in \Gamma_\infty$ and let $\dot{p}, \dot{x}, \dot{z}$ be $\mathbb{P}_P$ -names. There is one appended coding presentation in Γ_∞ which selects a preservation witness when $Y_P^\infty \neq \emptyset$, and otherwise codes $\dot{z}$.

Proof. In $\text{RO}(\mathbb{P}_P)$ put

$$b_\alpha = \bigvee \{a : \exists \dot{y} \text{ Pres}_\alpha(P, a; \dot{p}, \dot{x}, \dot{y}, m)\}.$$

Then $b_\alpha \leq b_\beta$ for $\alpha < \beta$. Partition 1 by

$$c_\rho = b_\rho \wedge \neg \bigvee_{\xi < \rho} b_\xi \quad (\rho \leq \theta), \quad c_* = \neg b_\theta.$$

On each nonzero c_ρ choose a maximal antichain of conditions a carrying names $\dot{y}_a$ with $\text{Pres}_\rho(P, a; \dot{p}, \dot{x}, \dot{y}_a, m)$. On c_* use $\dot{z}$. Mix these real names to obtain $\dot{w}$, and append $\text{Code}(\dot{p}, \dot{x}, \dot{w}, m)$ with origin P .

For $\alpha < \rho$, a condition in the c_ρ piece forces $Y_P^\alpha = \emptyset$. For $\rho \leq \alpha$ its selected name is protected at level α , since $\Gamma_\alpha \subseteq \Gamma_\rho$. On c_* every test is empty. Thus the same appended presentation satisfies every successor rule, and induction puts it in Γ_∞ . The pieces c_ρ include limit ranks; continuity of the preservation sets is not assumed. This mixes names for one coding stage, rather than tails of different lengths. $\square$

Lemma 4.13. At an admitted origin P , on the cone c_* where $Y_P^\infty = \emptyset$, each specified old candidate has a dense family of eliminating origin conditions. Scheduling its fixed binary continuations eliminates that candidate, with every countable initial segment admitted.

Proof. Fix $\dot{z}$. For every $0 < a \leq c_*$, failure of $\text{Pres}_\infty(P, a; \dot{p}, \dot{x}, \dot{z}, m)$ supplies an admitted R , a permitted origin-cone map and a stronger target condition r such that

$$r \Vdash_R \neg \varphi_m(j\dot{x}, j\dot{z}, j\dot{p}).$$

Cone transport gives a nonzero origin projection below a . Refine in the dense presentation of Lemma 4.8. Choose an actual origin condition $b_a \leq a$ and a corresponding fixed (R_a, r_a) whose projection contains b_a . Thus

$$D = \{b_a : 0 < a \leq c_*, a \in \mathbb{P}_P\}$$

is dense below c_* and $|D| \leq \omega_1$.

Schedule each (b_a, R_a, r_a) separately. Over the whole current history, activate its binary continuation when $b_a \in G_P$, and pad otherwise. Lemma 4.9 admits every such block. A generic in c_* meets D , so one scheduled block forces the negation. Its Σ_3^1 negation persists.

Each block has its own fixed length $\delta_a < \omega_1$. A countable initial agenda has length $\sum_{n < \omega} \delta_{a_n} < \omega_1$. We never assemble all the δ_a over one antichain into a single countable block. If a guess or a larger history has already been added, use the same origin and amalgamate with that whole history. $\square$

We shall use the following standard absoluteness observation without further comment. If $M \subseteq N$ are transitive models of ZFC with the same ordinals, N is a forcing extension of M , and $a \in M$ is a real parameter, then every $\Sigma_3^1(a)$ statement true in M remains true in N ; equivalently, every $\Pi_3^1(a)$ statement true in N was already true in M . This follows from Shoenfield absoluteness after writing the statement as $\exists u \psi(u, a)$ with ψ of complexity Π_2^1 .

Lemma 4.14. *Let $P \in \Gamma_\infty$ and let G be generic. Every entry in I satisfies $\varphi_m(x, y, p)$ in $W[G]$. Two distinct preservation witnesses for the same (m, p, x) certify that $A_{m,p}$ is not a graph. Consequently, if it is a graph, at most one value occurs in such preservation entries.*

Proof. The origin certificate quantifies over the admitted ambient presentation and its origin embedding. Apply the forcing relation at that embedding. The remaining assertions follow by applying this to both witnesses. $\square$

Remark 4.15. No ban on competing preservation values is imposed. Two amalgamated branches may record two values and show that the relation is not a graph. A Σ_3^1 -uniformization rule that keeps the first witness and bans competing codes is not closed under these amalgams.

Lemma 4.16. *In an ω_1 -length construction of these presentations, every final real belongs to a countable initial segment.*

Proof. Write the forcing as $C * \dot{Q}$, where C is the inherited W -product of the fixed reservoir cones and Q is the finite-support coding iteration. The coordinatewise reservoir isomorphisms show that C is σ -closed in W , also when its fixed root array has length ω_1 . Hence $W[G_C]$ has no new countable sequences of ordinals.

In that model, Q has dense conditions represented by finite arrays of coordinate ordinals, finite stems and finite restraint indices. Decide activity and cone-condition names by strengthening preceding coordinates; the finite dependency recursion terminates as in Lemma 4.8. Since Q is ccc, a nice name for a

real is a countable list of these finite arrays. This list has a countable ordinal code, so its particular code belongs to W .

Choose $\delta < \omega_1$ above its coding coordinates. The factors there use only their predecessor histories and their own reservoirs. Thus these arrays have the same order in the coding prefix over $W[G_{C,\delta}]$ as over $W[G_C]$. Its inherited prefix is regular, so the nice-name antichains are maximal there as well. Evaluating that same code with the prefix coding generic produces the given real in the original countable prefix. The choice of this code may depend on G_C ; no uniform syntactic bound for all real names is asserted. $\square$

Lemma 4.17. *Let $\mathbb{P}_{\omega_1}$ have countable initial segments in Γ_∞ . Every preservation entry in I satisfies its recorded $\varphi_m(x, y, p)$ in the final model. Every pair in D still certifies non-functionality there.*

Proof. Apply Lemma 4.14 in every countable later prefix. A final negation is Σ_3^1 and has a real witness u . By Lemma 4.16, u belongs to a countable later prefix. Shoenfield absoluteness gives the same negation there, a contradiction. Apply this to both witnesses in D . $\square$

5 The witnessing universe

We run an ω_1 -length construction over W , with countable initial segments in Γ_∞ . This final forcing is not itself a member of the countable-length class.

At each countable stage choose a name for an ω_1 -enumeration of the reals in that extension. Bookkeeping lists the indices for p, x, z, m cofinally often. It uses an admitted current prefix as origin, containing the supports of these names. Thus every required evaluated tuple is revisited. The records I, J, D include actions inside every appended tail. Each prefix creates at most ω_1 indexed real-data tasks, eliminating conditions and record-pair tasks. Dovetail these agendas, repeating each task cofinally often. A countable initial agenda contains only countably many fixed blocks. Consequently the total skeleton has length ω_1 . This lists evaluated reals by ordinal indices, rather than all raw names.

- (A) **Preservation action.** If $Y_P^\infty \neq \emptyset$, use the rank selection of Lemma 4.12 and code the selected y . Record its origin and certificate in I . For each earlier guessing entry with the same (m, p, x) , schedule the fixed squaring tasks below and record their two witnesses in D .
- (B) **Guessing-destruction action.** If $Y_P^\infty = \emptyset$, code the bookkeeping value z and record the empty-origin action in J . Schedule the dense family of fixed destruction blocks from Lemma 4.13. Their quotients are amalgamated over this origin with the whole current history.

Lemma 5.1. *Every countable initial segment of the final construction belongs to Γ_∞ .*

Proof. Coding uses Lemma 4.12. Destruction and squaring use fixed binary blocks, and countable limits use Lemma 4.9. Every choice is a fixed W -task; only its activation is a name. Thus no unbounded antichain assembly is needed. $\square$

Lemma 5.2. *Let*

$$j : \text{RO}(\mathbb{P}_P) \restriction c \longrightarrow \text{RO}(\mathbb{P}_Q) \restriction q_0$$

be an admitted origin-cone embedding. On its active cone, if $Y_{P, G_P, \dot{x}, \dot{p}, m}^\infty = \emptyset$, no real of $W[G_P]$ becomes a preservation witness at Q for the transported data.

Proof. Fix an old name $\dot{y}$ and put

$$d = \bigvee \{a : \text{Pres}_\infty(P, a; \dot{p}, \dot{x}, \dot{y}, m)\}.$$

If $t \leq q_0$ certifies the transported value at Q , then $t \leq j(d \wedge c)$. Otherwise

$$t' = t \wedge \neg j(d \wedge c) > 0.$$

Choose an origin condition $0 < a \leq \pi_j(t')$. Then $a \leq c \wedge \neg d$, so a is not a certificate. Its failed test supplies an admitted R , a permitted map on an origin subcone, and a stronger condition r forcing the negation. Cone transport gives its nonzero origin projection $b \leq a$. Now $t' \wedge j(b) > 0$. Amalgamate this cone and $R \restriction r$ over $P \restriction b$. The transported Q -certificate gives membership; the Σ_3^1 negation on the R -branch persists, a contradiction.

Thus a generic containing a later certificate contains $j(d \wedge c)$. The induced P -generic meets a maximal antichain of old certificates below d , contrary to $Y_P^\infty = \emptyset$. If the later name merely equals an old real in the generic extension, strengthen first to force that equality. All assertions are relative to the active cone q_0 . $\square$

Lemma 5.3. *If $Y_{P, G_P, \dot{x}, \dot{p}, m}^\infty = \emptyset$, there is an ω_1 -length inherited construction which destroys every candidate in $W[G_P] \cap \mathbb{R}$, with all countable stages admitted.*

Proof. Enumerate the old reals using CH. For each indexed candidate, schedule the dense family in Lemma 4.13. Apply its fixed binary blocks over the entire current history and use Lemma 4.9 at countable limits. Each Σ_3^1 negation persists through the remaining construction. $\square$

We also use the product-name separation principle. For independent copies of a forcing over M , a product condition forcing two copied real names equal already forces them to be one real of M . Hence, if one interpretation is new over M , the second copy forces a distinct interpretation; cf. [19].

Lemma 5.4. *Let $P \restriction c \rightarrow Q \restriction q$ be an admitted origin-cone embedding. A copy Q^* fixing this P -cone retains all original names and records, and*

$$(Q \restriction q) \otimes_{P \restriction c} (Q^* \restriction q^*) \in \Gamma_\infty.$$

The branches have the same names for parameters from P .

Proof. Use Lemma 4.9. Relabel only coordinates outside the common cone. The copied record has the same abstract origin and test; its current map is composed with the copied branch inclusion. Condition restrictions retain dormant records as well. $\square$

Lemma 5.5. *Suppose an earlier origin P has an empty preservation set for (m, p, x) , and a later admitted history Q records a preservation witness y . Scheduled fixed binary continuations force that $A_{m,p}$ is not a graph.*

Proof. Put $M = W[G_P]$. Lemma 5.2 gives $y \notin M$. The origin certificate lifts to the current active Q -cone: restrict its origin map to each tested subcone and compose with the continuation map. For these two fixed records, list the decided Q -conditions q activating them, equating their data, and forcing that the later value is new over M . Some such q belongs to the actual generic. There are at most ω_1 record-pair and condition tasks in a countable Q . Newness is tested against the range of the fixed P -name enumerating its reals, so this is a W -forcing assertion.

The q -task uses one fixed copy of $Q \restriction q$, fixing the earlier P -cone and relabeling coordinates outside P . Over the whole current history U , guard it by $d = j_U(q)$. On d append its copied quotient; on $\neg d$ pad with the same fixed reservoir roots and length. This is Lemma 4.8.

In the two branches, a product condition forcing the copied real names equal would force their common value to belong to M , contrary to the choice of q . Hence their values y_0, y_1 are distinct. The transported certificates give

$$\varphi_m(x, y_0, p) \wedge \varphi_m(x, y_1, p) \wedge y_0 \neq y_1.$$

Both certificates retain their original origins. Record them in D ; Lemma 4.17 preserves them to the final model. $\square$

Remark 5.6. A guessing entry records an origin with an empty preservation set. It does not assert membership of the guess in the section. Any later preservation witness is new over that origin and can be squared. This applies also to guessing entries inside destruction or copied tails; those entries need not have destroyed their own guessed values.

Let G_{ω_1} be generic and put $W^* = W[G_{\omega_1}]$. Every real appears in a countable initial segment. Close its support under the countable dependency closure and use exhaustive bookkeeping as above.

Lemma 5.7. *In W^* , if $A_{m,p}$ is the graph of a total function, then*

$$\Phi(p, x, y, m) \iff \varphi_m(x, y, p).$$

Consequently $A_{m,p}$ is $\Sigma_3^1(p)$.

Proof. Let f be the total function. Suppose a guessing entry for (m, p, x) was recorded at origin P with $Y_P^\infty = \emptyset$, and put $y = f(x)$. Choose a later countable prefix containing this origin and names for p, x, y . Bookkeeping presents y as a

candidate over a later full prefix. If its preservation set were empty, destruction would make $\neg\varphi_m(x, y, p)$ persist to W^* , a contradiction. Thus a preservation action records some y_0 . Final persistence and functionality give $y_0 = y$. Non-promotion and squaring then give two distinct values in the section, again a contradiction. Hence no stable guessing entry occurs for this section. This argument includes entries introduced inside appended tails.

If $\Phi(p, x, w, m)$ holds, Lemma 3.4 gives a first explicit coding stage. Its stable classification cannot be guessing, so it has a preservation certificate. Lemma 4.17 gives $\varphi_m(x, w, p)$.

Conversely, if $\varphi_m(x, y, p)$ holds in W^* , bookkeeping eventually presents y over a prefix containing its name. The empty-set action would destroy it, so a preservation witness y_0 is selected. Final persistence and functionality imply $y_0 = y$. Its code is introduced, and persists by Lemma 3.4. $\square$

Theorem 5.8. *In W^* every total Π_3^1 -function has a Σ_3^1 graph.*

Proof. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a total function in W^* whose graph is Π_3^1 . Choose $m < \omega$ and a real parameter p such that

$$\text{Graph}(f) = A_{m,p} = \{(x, y) : \varphi_m(x, y, p)\}.$$

By Lemma 5.7, for all reals x, y in W^* ,

$$(x, y) \in \text{Graph}(f) \iff \Phi(p, x, y, m).$$

Since Φ is uniformly Σ_3^1 in the parameter p , the graph of f is $\Sigma_3^1(p)$. $\square$

6 Combining with Π_3^1 -reduction

We finally explain how the preceding construction is combined with the forcing construction for Π_3^1 -reduction from [6]. We use only the fixed point construction for reduction and its closure properties. The erroneous assertion is Lemma 3.13 in the last part of [6], concerning graphs of partial Π_3^1 functions. This lemma in fact does not hold in $W[G_{\omega_1}]$, contrary to what the lemma claims¹. It is replaced by the total-graph argument in Sections 4 and 5.

Fix, in addition to the enumeration $\langle \varphi_m(v_0, v_1, v_2) \mid m < \omega \rangle$ of binary Π_3^1 formulas used above, a recursive enumeration

$$\langle \theta_m(u, v) \mid m < \omega \rangle$$

of all lightface Π_3^1 formulas in two real variables, chosen so that

$$\theta_0(u, v) \equiv u = v.$$

¹Indeed, the proof erroneously claims that under the lemmas assumptions case 1 (i) applies in the iteration. A short look at the definition reveals that in fact case 1 (ii) must apply which implies that lemma 3.13 is not true as stated. The intuition to consider Π_3^1 graphs of functions and their Σ_3^1 -definability was the right one though, as this section shows. The lemma 3.13 can never hold as it would collapse the projective hierarchy.

For a real parameter p , write

$$B_{m,p} = \{x \in \mathbb{R} \mid \theta_m(x, p)\}.$$

Thus every boldface Π_3^1 set of reals is some $B_{m,p}$.

We refine the fixed definable sequence of coding trees into three definable pairwise disjoint subsequences

$$\vec{S}^{\text{gr}}, \quad \vec{S}^{\text{red},1}, \quad \vec{S}^{\text{red},2}.$$

This is just a definable thinning of $\vec{S}$. The graph construction of the previous sections is now read as using only $\vec{S}^{\text{gr}}$; its decoding predicate will be denoted by

$$\Phi^{\text{gr}}(p, x, y, m).$$

The reduction construction uses the two sequences $\vec{S}^{\text{red},1}$ and $\vec{S}^{\text{red},2}$. We write

$$\Psi_i(p, x, m, k) \quad (i = 1, 2)$$

for the usual Σ_3^1 statement that the real coding (p, x, m, k) has been written into the i -th reduction sequence. The three tree subsequences and the forcing-coordinate families are chosen disjointly. Within each subsequence the reservoir coding areas are almost disjoint, as in Section 3.

We use the first-preservation-rank rule of [6, Definition 3.5], with the origins and cones of Section 4. The hierarchy is computed once in W . Both kinds of preservation tests quantify over the combined class.

Definition 6.1. *Let Γ_0^{cmb} consist of the normal forms of Section 4, using the three coding families above. For an admitted origin P and $\mathbb{P}_P$ -names $\dot{p}, \dot{x}$, put*

$$d_{P,\alpha}^1 = \bigvee \{a : \text{Pres}_\alpha^1(P, a; \dot{p}, \dot{x}, m, k)\}, \quad d_{P,\alpha}^2 = \bigvee \{a : \text{Pres}_\alpha^2(P, a; \dot{p}, \dot{x}, m, k)\}.$$

Here Pres_α^1 is Definition 4.2 for $\theta_m(\dot{x}, \dot{p})$, and Pres_α^2 is the same test for $\theta_k(\dot{x}, \dot{p})$, using $\Gamma_\alpha^{\text{cmb}}$. All joins are in $\text{RO}(\mathbb{P}_P)$. Write

$$b_{P,\alpha} = d_{P,\alpha}^1 \vee d_{P,\alpha}^2, \quad c_{P,\rho} = b_{P,\rho} \wedge \bigwedge_{\xi < \rho} b_{P,\xi}.$$

The first-rank choice through level α is the name $i_P^\alpha \in \{0, 1, 2\}$ given by

Boolean piece	i_P^α
$c_{P,\rho} \wedge d_{P,\rho}^1 \quad (\rho \leq \alpha)$	1
$c_{P,\rho} \wedge \neg d_{P,\rho}^1 \quad (\rho \leq \alpha)$	2
$\neg b_{P,\alpha}$	0.

Thus side 1 has priority if both memberships are protected at the first rank. Protection appearing at a later rank does not change that choice.

A presentation belongs to $\Gamma_{\alpha+1}^{\text{cmb}}$ if it belongs to $\Gamma_\alpha^{\text{cmb}}$, its retained origins belong to that class, and its actions satisfy the following rules on their recorded origin conditions.

- (a) **Reduction requirement.** For data (p, x, m, k) , with $m, k \neq 0$, a code written into reduction sequence $i \in \{1, 2\}$ is permitted precisely when

$$a \Vdash_{\mathbb{P}_P} i_P^\alpha = 0 \vee i = 3 - i_P^\alpha.$$

The selected sequence may be a name read at the origin. The original Y -recipe then uses that sequence and the real coding (p, x, m, k) .

- (b) **Total-graph requirement.** Use Definition 4.3, with the combined preservation class and the graph coding family. Preserve the original origin and names.

At a limit take the intersection of the same decorated presentations. Padding has no test. Later continuations are inherited quotients; no derivative is recomputed in a generic extension.

Lemma 6.2. Let $P, Q \in \Gamma_\alpha^{\text{cmb}}$ and let

$$j : \text{RO}(\mathbb{P}_P) \restriction c \longrightarrow \text{RO}(\mathbb{P}_Q) \restriction q$$

be a permitted origin-cone embedding. For the transported datum,

$$d_{Q,\alpha}^i \wedge q = j(d_{P,\alpha}^i \wedge c) \quad (i = 1, 2). \quad (*)$$

Consequently first ranks and selected sides agree on the corresponding cones.

Proof. If a is a certificate at P , consider any nonzero $q' \leq j(a \wedge c)$. Put $c' = \pi_j(q') \leq a \wedge c$ and restrict the origin map to

$$j_{q'} : \text{RO}(\mathbb{P}_P) \restriction c' \longrightarrow \text{RO}(\mathbb{P}_Q) \restriction q', \quad j_{q'}(u) = j(u) \wedge q'.$$

Compose any permitted continuation of this Q -cone with $j_{q'}$. The original P -certificate applies to the composite. Thus the image of $a \wedge c$ is a Q -certificate. Taking joins gives the right-to-left inclusion in $(*)$.

For the reverse inclusion, suppose a Q -certificate has a nonzero part

$$q' \leq q \wedge \neg j(d_{P,\alpha}^i \wedge c).$$

Choose an origin condition $0 < a \leq \pi_j(q')$. Then $a \leq c \wedge \neg d_{P,\alpha}^i$, so a is not a certificate. Failure of its preservation test supplies an admitted extension R , a permitted origin-cone map, and a condition r forcing the negation of the i -th membership. Restrict this map to r ; its domain is a nonzero cone $b \leq a$. Since $b \leq \pi_j(q')$, one has $q' \wedge j(b) > 0$. Amalgamate this Q -cone and the R -cone below r over their common origin cone below b . The Q -certificate gives the membership there. The negation forced on the R -branch is Σ_3^1 , so it persists to the amalgam. This is a contradiction. The argument is the no-promotion proof at level α . The closure induction of Lemma 4.9 applies to the combined rules, as verified below; its origin-cone maps give the amalgam used here. Apply $(*)$ at every earlier level to obtain the last assertion. If a condition equates two sets of parameter names, their protection predicates agree below that condition: every permitted cone map transports the forced equality. The same is true of their first-rank tables. $\square$

Lemma 6.3. *The combined hierarchy is definable, monotone, nonempty, and closed under the normal-form operations of Lemma 4.9. It stabilizes in W . Its stable class, denoted by $\Gamma_\infty^{\text{cmb}}$, admits one-stage coding extensions for every graph and reduction datum and the fixed-root binary destruction blocks used below.*

Proof. Use the induction of Lemma 4.9. Every retained action keeps its original origin condition, names and test. The first-rank rule is invariant under the permitted relabelings. In a cone restriction or binary mixture, an inactive action is padding; an active action has the same origin test as before. At a limit use the same normal form at every rank. This proves closure. The trivial and padding presentations prove nonemptiness.

The classes are decreasing subsets of the fixed set of normalized codes. If two consecutive classes agree, their preservation predicates agree too. No new first-rank piece is then added, so the successor rule is unchanged at every later level. Hence the hierarchy stabilizes. Let θ be its first stabilization ordinal.

For a reduction datum at a stable origin P , mix the names for the sequence i over

$$c_{P,\rho} \wedge d_{P,\rho}^1, \quad c_{P,\rho} \wedge \neg d_{P,\rho}^1 \quad (\rho \leq \theta), \quad \neg b_{P,\theta}.$$

On the first piece choose $i = 2$, on the second choose $i = 1$, and on the last choose either bookkeeping guess. Append one fresh reservoir and one coding factor, retaining P as origin. Before ρ both sides are unprotected; from ρ onward the first-rank rule requires exactly the selected sequence. Thus this same one-stage extension belongs to every level. Graph coding uses Lemma 4.12.

If a reduction membership has no certificate on a cone, failure of its preservation test gives eliminating cones with dense origin projections. For each fixed witness use the binary construction of Lemma 4.9, with the same fixed W -reservoir roots on its active and padding branches. Each block has its own fixed countable length. These blocks are scheduled separately; no collection of tails of unbounded countable lengths is assembled at one stage. $\square$

We now run an ω_1 -length construction over W . Bookkeeping handles both kinds of data and their scheduled continuations as in Section 5. At each countable prefix choose names for an ω_1 -enumeration of its reals and use ordinal indices. All records inside appended tails are retained.

At a reduction origin P , perform the coding extension just described. On $b_{P,\infty}$ it retains side i_P^∞ and codes the other side. On

$$c_* = \neg b_{P,\infty}$$

schedule destruction of both memberships. For $i = 1, 2$ choose in W a dense family $\mathcal{D}_i$ of origin conditions below c_* and one eliminating witness (R_b, r_b) for each $b \in \mathcal{D}_i$. There are at most $\aleph_1$ such conditions. Schedule every tuple

$$(P, p, x, m, k, i, b, R_b, r_b)$$

cofinally often. Its block amalgamates the entire current history with $R_b \upharpoonright r_b$ on the b -cone and pads the complementary cone. Use the fixed reservoir roots

of that one witness on both branches. If a membership is already false, its negation persists and needs no further action. Destruction of both sides is scheduled whenever the protection union is empty, even if only one membership currently holds.

At graph stages use the graph coding, destruction and squaring agendas of Section 5, with the combined hierarchy. Every chosen block has fixed countable length. A countable initial agenda therefore has countable ordinal sum. Let $G_{\omega_1}^{\text{cmb}}$ be generic and put

$$W^{\text{cmb}} = W[G_{\omega_1}^{\text{cmb}}].$$

Lemma 6.4. *Every countable initial segment of this construction belongs to $\Gamma_{\infty}^{\text{cmb}}$. The final forcing preserves ω_1 and satisfies CH.*

Proof. Coding extensions and each scheduled binary block belong to the stable class by Lemma 6.3. Squaring uses a copied branch and common-prefix amalgamation. Countable concatenations use Lemma 4.9. The underlying normal forms and their ordinary envelopes give preservation and CH as in Lemma 3.4. $\square$

Lemma 6.5. *In W^{cmb} , the Π_3^1 -reduction property holds.*

Proof. Fix p, m, k with $m, k \neq 0$ and write

$$D_{m,k,p}^1 = B_{m,p} \cap \{x : \neg \Psi_1(p, x, m, k)\}, \quad D_{m,k,p}^2 = B_{k,p} \cap \{x : \neg \Psi_2(p, x, m, k)\}.$$

These sets are $\Pi_3^1(p)$.

Let $x \in B_{m,p} \cup B_{k,p}$ in the final model. Bookkeeping considers the datum at a countable origin P . Its generic cannot lie in $\neg b_{P,\infty}$: it would meet both dense eliminating families, and their scheduled blocks would make both negations persist to the final model. Hence the coding action retains a side $s \in \{1, 2\}$ and codes the other side. Its certificate gives final membership on side s by the argument of Lemma 4.17.

No code for this datum is written into sequence s . Indeed, compare any such coding origin A with P inside a later full history Q , restricting to their active cones and to a condition forcing the data equal. By (*), an empty protection union at A would remain empty at Q , contrary to the certificate from P . Otherwise the whole first-rank table transports from both origins to Q , so the selected sides agree. The action at A must therefore code the other sequence. This applies also to codes inside destruction and copied tails.

The exact decoding lemma for coned normal forms, obtained from their ordinary envelopes in Section 4, now gives $\neg \Psi_s(p, x, m, k)$. Thus x belongs to the retained reducing set. If x belongs to the intersection of the two original sets, the code into the other sequence removes it from that reducing set. This proves coverage and disjointness. If one index is 0, one original set is all of $\mathbb{R}$ and reduction is immediate. Two different real parameters may be joined into one real parameter. $\square$

Lemma 6.6. *In W^{cmb} , if $A_{m,p}$ is the graph of a total function, then*

$$A_{m,p}(x, y) \iff \Phi^{\text{gr}}(p, x, y, m).$$

In particular, every total Π_3^1 graph is Σ_3^1 .

Proof. Repeat Lemma 5.7 with the combined hierarchy. Its preservation tests include reduction actions. Closure, rank selection, scheduled destruction and common-prefix squaring hold by Lemma 6.3. A final value cannot lie at an empty preservation origin, since its dense elimination agenda would destroy it. A later preservation action selects that value by functionality. An earlier guessing marker then gives newness and squaring, a contradiction. Hence every graph code has a preservation certificate, and every final value is coded. Exact decoding for the coned envelope gives the equivalence; disjoint coordinate labels alone are not used to infer it. $\square$

Theorem 6.7. *Assuming $\text{Con}(\text{ZFC})$, it is consistent that Π_3^1 -reduction holds, Π_3^1 -uniformization fails, and every total Π_3^1 -function has a Σ_3^1 graph.*

Proof. Lemma 6.4 gives the forcing construction. Lemmas 6.5 and 6.6 give reduction and the total-graph property. Corollary 2.2 gives failure of uniformization. $\square$

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